How do I determine which printer I need?

If you run a small shop, pharmacy, or coffee shop and only print simple price tags or box stickers, a desktop thermal printer is the right choice. It is compact, easy to use and works without a ribbon, only on thermal paper. Drawback: labels fade over time (typically within 3–6 months).

If you work with a warehouse, logistics, or production where thousands of labels must be printed per day and durability matters, choose a thermal transfer printer. It uses a ribbon, so labels do not rub off or peel and withstand cold, moisture and sunlight.

If you are required to print marking (DataMatrix), e.g., for pharmacies, apparel or tobacco products, you definitely need a high-resolution thermal transfer printer (300 dpi or higher). Only such a device provides a crisp, readable code accepted by the system.

Summary: for small volumes and simple tasks — pick a desktop thermal printer. If you need headroom or print marking — pick thermal transfer. If unsure, we'll suggest the optimal model so you don't overpay.

What is the difference between a thermal transfer printer and a thermal (direct thermal) printer?

A thermal (direct thermal) printer prints directly on thermal paper. No ribbon is needed — only label rolls. This is convenient when you print:

- price tags in a store;
- scale labels;
- stickers with a short service life (from a few weeks to a couple of months).

The downside is that such labels fade quickly, can smear, and are sensitive to moisture and sunlight. If goods are stored in a refrigerator or outdoors, labels may peel off or become unreadable.

A thermal transfer printer works with a ribbon (inked tape). The print is more durable and suitable for:

- product marking (DataMatrix codes);
- warehouses and logistics;
- pharmacies where barcode readability is critical;
- production environments where labels must withstand cold, chemicals and abrasion.

The downside is the need to purchase a ribbon. However, such labels last for years and do not rub off.

Summary: for simple and low-cost printing — use a direct thermal printer. If reliability and durability (especially marking) are required — choose thermal transfer.

Can a single printer cover all label printing tasks?

Unfortunately, there is no "magic universal printer" — the choice depends on your tasks. Still, we can pick an option that fits your specific needs.

For a small store or pharmacy where you need simple price tags or stickers, a desktop thermal printer is sufficient. It is compact, inexpensive and handles up to several thousand labels per month.

If you run a warehouse or production printing thousands of labels daily, an industrial thermal transfer printer is preferable. It is built for heavy loads and durable output.

If you must apply mandatory marking (DataMatrix), there is no alternative: you need a high-resolution (300 dpi or higher) thermal transfer printer. Only it provides a quality code that scans correctly.

Summary: you can select a printer that covers your core tasks, but there is no one-size-fits-all device for every business. Always base the choice on print volumes and legal requirements.

How do industrial printers differ from desktop models?

Desktop printer:

- Compact, lightweight and space-saving.
- Designed for smaller volumes — from hundreds up to a couple of thousand labels per day.
- Commonly used in stores, pharmacies, small warehouses and offices.
- Works mainly with paper and thermal paper, suitable for price tags, barcodes and simple labels.
- Easy to operate: no special skills required; connects to a computer or POS easily.

Industrial printer:

- Larger size, reinforced construction and resistant to continuous loads.
- Can run around the clock and print tens of thousands of labels daily.
- Used in warehouses, logistics centers and production lines.
- Supports various materials: paper, synthetics, film — for refrigeration, sun exposure or transport conditions.
- Equipped with advanced features (high print speed, large media rolls, integration with accounting/ERP systems).

Summary: desktop printers are for smaller volumes and simple tasks. Industrial models are professional equipment for high throughput, varied materials and prolonged continuous operation.

How to choose a self-adhesive label by material?

Thermal labels — paper labels with a heat-sensitive layer printed without a ribbon. Used in retail, pharmacies, at POS and in logistics where product life is short. Pros: low cost, convenience, no extra consumables. Cons: fade quickly; sensitive to moisture, sunlight and abrasion.
Semi-gloss paper — smooth paper labels printed with a ribbon. Suitable for food, cosmetics, household chemicals and other goods where appearance matters. Pros: versatility, neat and vivid result. Cons: lower resistance to moisture and mechanical stress.
Matte paper — rougher paper labels printed via ribbon. Used for warehouse accounting, boxes/packaging and internal markings. Pros: economical and simple; con: less premium look and lower contrast than semi-gloss.
Film (“pearl”) — synthetic, durable labels that withstand moisture, sun and temperature swings. Used in freezers, warehouses, logistics and on goods stored or transported for long periods. Pros: maximum durability; con: higher price than paper.
Summary: for quick and inexpensive printing on short-life goods — thermal labels. If appearance matters — choose semi-gloss. For budget warehouse tasks — matte paper. For durability and environmental resistance — film (“pearl”).

How to choose a ribbon for printing?

Wax — a film with wax-based ink. During printing the wax melts and transfers to the label surface. Works well on paper stocks (semi-gloss, matte). Pros: low cost, excellent barcode/QR quality. Cons: wax rubs off easily — sensitive to moisture, abrasion and high temperatures.
Wax/Resin — a ribbon where wax is blended with resin. Thanks to resin, the print is more durable and lasts longer than pure wax. Suitable for paper (semi-gloss, matte) and some synthetics. Pros: balance of price and durability; versatile. Cons: more expensive than wax yet not as long-lasting as pure resin.
Resin — a film with synthetic resin-based ink. The resin effectively “fuses” into the label surface. Used for synthetic materials (film, “pearl”) where maximum durability is required. Pros: resistant to moisture, cold, chemicals and abrasion. Cons: the most expensive option and requires precise printer settings.
Summary: no ribbon is used for thermal labels. For paper materials use Wax; for greater durability use Wax/Resin. For synthetics and harsh conditions, Resin is optimal.

How do I know whether I need a ribbon or can do without it?

If your tasks involve short storage periods and the label does not need to last long, you can safely use thermal labels without a ribbon. This suits POS receipts, price tags, logistics stickers, pharmacies and food delivery — where the label “lives” from a few days to a couple of months.
If your goods are stored longer, frequently transported or exposed to the elements (moisture, cold, abrasion), use labels printed with a ribbon. Such labels (semi-gloss, matte paper, film “pearl”) are more reliable: barcodes and DataMatrix remain readable and do not fade. This is important for production, warehouses, legally mandated marking and products with long shelf life.
Summary: for simple, low-cost “here and now” labels — thermal labels without a ribbon. If durability, reliability and compliance with marking requirements matter — use thermal transfer materials with a ribbon.

Which labels are best for storage in a refrigerator or freezer?

For cold and humid conditions, both the label stock and adhesive must withstand low temperatures and condensation.
Thermal labels — not suitable: in the cold they darken quickly and smear, especially in the presence of moisture.
Semi-gloss paper — acceptable if paired with a Wax/Resin ribbon. But in a freezer, paper may soften if it contacts moisture.
Matte paper — behaves similarly to semi-gloss: fine for a refrigerator, risky for a freezer.
Film (“pearl”) — the best option. A synthetic label resists moisture, cold and sub-zero temperatures. With a Resin ribbon, the print will be resistant to frost and abrasion.
Summary:
For refrigerators (0…+5 °C) — semi-gloss or matte paper with Wax/Resin.
For freezers (–18 °C and below) — film + Resin. This solution won’t peel off and won’t smear even with condensation.

What should I do if the printer prints poorly or the ribbon smears?

Possible causes:
- Ribbon not matched to the label material.

Use Wax or Wax/Resin for paper; Resin for film. If the media and ribbon don’t match, prints will smear.
- Incorrect print temperature.

If too low — the print is faint; if too high — the ribbon can smear.
- Dirty printhead.

Adhesive and dust accumulate over time, so clean it regularly. Ideally at each ribbon change.
- Consumable quality.

Very cheap materials may simply not hold the print properly.
Summary: first adjust temperature, then clean the printhead. If that doesn’t help, the issue is likely the consumables — replace them.

Can I print labels with my company logo or in color?

Yes. There are several options:

Black-and-white printing on thermal or thermal transfer printers.

✓ You can add a logo to the label layout.
✓ Printing will be in black only.
✓ Perfect for barcodes and simple logos.

Color ribbons.

✓ Replace the standard black ribbon with blue, red, green, gold, etc.
✓ Highlights a logo or important text.
✓ Multi-color (photo-like) printing is not possible — one color per ribbon roll.

Color printing on dedicated printers.

✓ Typically inkjet or laser models.
✓ Enables full-color labels (branded, promotional, with images).
✓ Downside — higher print cost.

Pre-printed color labels + overprint.

✓ We can supply labels with a color background and logo in advance.
✓ Your printer then prints only barcodes, dates or ingredients.
✓ The most budget-friendly way to add color.

Summary: labels with a logo or color elements are absolutely possible. For simple highlighting, a color ribbon is enough; for branded products, use pre-printed labels or a color printer. We'll help select the best option for your business.

What if the printer doesn't connect to the computer?

Check power and cables.
The printer must be on and indicators lit.
Power and USB/network cables must be firmly inserted.
If that doesn't help, try another USB port or cable.
Check whether the printer prints by itself.
Most models have a "Feed" or "Pause" button. Hold it to print a test label.
If the test prints, the printer is fine — the issue is on the computer/settings side.
Check the computer.
Open "Printers & Scanners" in Windows or Mac. If the device is listed, remove and add it again.
If it is not listed, install the driver.
Install the driver.
Download it from your printer manufacturer's official website.
Choose the driver for your OS and install it.
Reboot the computer after installation.
If the printer is on the network (Wi-Fi or Ethernet).
Find the printer's IP address (often printed on the test label).
Ensure the computer and printer are in the same network.

Summary: start with cables and power, then print a test. If the printer works but the computer doesn't see it, install the driver. If that still fails, the issue is likely in settings or consumables. We provide support and will help resolve it.

Where do I download printer drivers?

Each printer works correctly only with the manufacturer's own driver.
Best practice — download the driver from the official website (Zebra, TSC, Godex, Honeywell, etc.).
Select your exact printer model on the site.
Download the driver for your operating system.
Install and reboot your computer.
Alternatively, a driver may come on a disk/USB with the device. Still, we recommend downloading the latest version from the official site.
Important: do not download drivers from untrusted sites — they may be outdated or even malicious.

Summary: always get drivers from the manufacturer's official website. If you cannot find the right driver or are unsure which one you need — contact us and we'll help install it.

Can I set up the printer myself or do I need a specialist?

Printer setup is usually straightforward; many clients handle it themselves.
You can typically do the following on your own:

- Install the driver.
- Connect the printer to the computer.
- Configure label size in the printing software.
- Select the correct print mode (direct thermal or thermal transfer with ribbon).

Call a specialist when:

- The printer outputs defects (smearing, double lines, label feed issues).
- You need integration with your accounting/ERP system (e.g., 1C).
- You cannot connect multiple printers to one network.

Summary: basic setup can be done yourself; for more complex integration or persistent errors, contact us — we will configure everything properly.

Is it hard to print barcodes and QR codes?

In fact, printing barcodes and QR codes is no harder than printing regular text.
Printing software does this automatically.
You only input the required numbers or text — the software converts them into a barcode or QR code.
All modern printers support it.
Any thermal or thermal transfer printer can print barcodes and QR codes without extra modules.
Choosing the right size is important.
If the code is too small, scanners may fail to read it.
If it is too large, it wastes label space.
Summary: printing codes is easy — the software handles it. Just set an appropriate size. If something goes wrong, our specialists will help.

Which software is best for printing labels?

There are several options, and the choice depends on your tasks:
Free manufacturer software.
Most brands provide their own tools (e.g., GoLabel, ZebraDesigner).
✓ Suitable for basic tasks: barcodes, QR, text, logo.
✓ Connects to the printer easily.
Specialized business software.
For example, BarTender or NiceLabel.
✓ Convenient for large volumes.
✓ Can connect databases to batch-print many labels with different data.
Simple office tools.
Sometimes Word or Excel are used, but they're inconvenient and limited — typically only for rare one-offs.
Summary: free tools bundled by the manufacturer are fine for simple tasks. For heavier volumes and automation, use professional solutions. We'll help you choose and configure the software.

How to care for a printer so it lasts longer?

A label printer is a workhorse; with proper care, it serves for years. Follow a few simple rules:
Clean the device regularly. After each ribbon change or every few rolls, wipe the printhead with an alcohol wipe and remove dust and paper debris inside the housing.
Store consumables properly. Do not expose label rolls and ribbons to sunlight or humidity; they degrade and can impair print quality. Very cheap materials also wear the printer faster.
Use the printer carefully. Media must load easily, without force. Do not print on unsuitable materials or overload the device beyond its design volume. Desktop models, for example, are for hundreds, not thousands, of labels a day.
Summary: cleaning, quality consumables and reasonable workloads keep the printer reliable for a long time. If issues arise, we can help with repair and maintenance.

Does a printer need regular maintenance?

Yes. Even if it runs smoothly, maintenance matters — like with a car. Regular checks and care help prevent major failures.
Periodic cleaning is beneficial — wipe the printhead, remove dust and adhesive residue. Paper scraps or traces of media may remain inside and eventually interfere with print quality.
Monitor the condition of pinch rollers and the printhead. They have a finite life and, when worn, cause lines or faint prints. If detected early, replacing a part is cheaper and faster than repairing the whole printer.
Summary: regular maintenance extends service life and prevents surprises. We recommend light preventive care every few months; for high print volumes, do it more often. If needed, we'll advise or perform the service for you.

How to clean the printhead and what to use?

The printhead is the most important and sensitive part of the printer. It forms the image on the label, and print quality directly depends on its condition.
Clean it regularly — ideally every time you change the ribbon or after several rolls. Use special alcohol wipes or a cotton swab lightly moistened with isopropyl alcohol.
Important: wipe gently, without pressure, so as not to damage the surface. After cleaning, let the printhead dry completely — usually a couple of minutes is enough — and then resume printing.
Summary: cleaning the printhead is simple and fast, and the result is immediate: crisper prints, less media wear and longer printer life. If you're unsure, we can demonstrate and train your staff.

How long does it take to set up a printer and start printing?

Setup usually takes little time. For new equipment, connecting, installing drivers and basic configuration takes around 20–30 minutes.
If you've worked with such a printer before, it's even faster — just connect it and select the required parameters in the software.
Summary: on average, printing can be launched within about half an hour. If needed, we can assist remotely or on site.

If a printer breaks, can it be repaired or is it easier to buy a new one?

Label printers are reliable devices; in many cases they can be repaired — replacing the printhead, roller or power supply. This is usually cheaper than buying a new printer.
If the model is very old or the repair cost approaches the price of a new device, purchasing a new printer may be more economical.
Summary: in most cases repair is feasible. We will assess the situation and advise whether repair or replacement is better.

What if the printing software doesn't see the printer?

Sometimes the application doesn't detect the printer even when it's connected. Possible reasons:

- The printer is not selected as the default device in the software.
- No connection to the computer (cable, Wi-Fi or Bluetooth).
- Drivers are not installed.
- The software and printer are set to different ports (e.g., USB vs COM).

What to do:

- First check whether the computer sees the printer (in "Devices and Printers").
- Make sure the driver is installed.
- Select the appropriate printer in the software.
- Reconnect the cable or restart the device.

Summary: most issues are solved by fixing the connection and selecting the correct printer in the app. If not, we will help connect and configure it.

Which goods in Uzbekistan are subject to mandatory digital marking?

As of today, the following product groups in Uzbekistan are subject to mandatory digital marking:

- tobacco products,
- alcoholic beverages,
- beer and beer-based drinks,
- medicinal products,
- water and soft drinks,
- household appliances.

Bottom line: if your business deals with any of these categories, marking is a legal requirement. For other goods, it is not yet required.

How to print DataMatrix codes correctly?

DataMatrix codes are special 2D codes used in the marking system and must be precise. They cannot be printed on a regular direct thermal printer meant for simple labels without a ribbon. Proper marking requires:

- Thermal transfer or specialized industrial printers with high resolution (typically 300 dpi or higher).
- Quality labels and ribbons (Wax/Resin or Resin) to ensure long-term readability.
- Quality control using a scanner before dispatch: the code must be readable on the first try, otherwise the system will reject it.
- Do not change the size, shape or elements of the code — even small distortions render it invalid for the system.

Bottom line: printing DataMatrix requires dedicated equipment, materials, quality control and strict adherence to standards. A regular thermal printer is not suitable — only the proper setup ensures acceptance of your goods.

Can I use a regular printer to print marking codes?

No. A regular office printer or a paper-only thermal printer is not suitable for product marking in the digital system. Reasons:

- Print quality. DataMatrix is a 2D code read by dedicated scanners. Any blurring or distortion makes the code unreadable.
- Materials. Regular printers do not work with thermal transfer ribbons and specialized labels required for durable marking.
- Service life. Even if printed on plain paper, the code will quickly wear off, get wet or peel, especially in refrigeration, freezing or during transport.

Bottom line: printing marking codes requires specialized thermal transfer or industrial printers with proper supplies. Regular office printers cannot be used — they do not meet system and legal requirements.

What happens if codes are printed incorrectly or if goods are not marked?

If DataMatrix codes are printed with errors, poorly or not according to the system standard, the goods:

- will not be accepted into circulation and cannot be legally sold;
- cannot be tracked in the system, violating legal requirements;
- expose the business to fines, since non-compliance is an administrative offense;
- risk returns or blocking, causing extra costs and sales delays.

If goods are not marked at all, consequences are even more serious:

- fines are significantly higher;
- goods may be confiscated upon sale attempts;
- the business risks losing trust of customers and partners.

Bottom line: correct and precise printing of codes is mandatory. This is not a formality — it is the law that protects the market. Violations directly lead to fines and problems with product circulation.

How to connect a printer to the “ASL Belgisi” marking system?

Registration in the “ASL Belgisi” system.

First, companies engaged in retail or wholesale of goods subject to mandatory digital marking must register in the national information system NIS “ASL Belgisi”.

Contracts.

After registration, two contracts are required:

- a connection agreement to the marking system — one for all product groups;
- an agreement for the provision of marking codes — separate for each product group.

After signing, you’ll get access to the personal account to manage marking processes. (crpt-turon.uz)

Selecting and configuring the printer.

To print DataMatrix codes, use thermal transfer printers with at least 300 dpi resolution. Regular office printers are not suitable.

Installing software.

You need software that can:

- generate DataMatrix codes;
- transmit product information to the “ASL Belgisi” system;
- integrate printing with the printer and your accounting/WMS (e.g., 1C or warehouse software).

Examples: GoLabel (free), BarTender (professional with extended features).

Printing and quality checks.

After configuring the printer and software, you can start printing. Regularly verify quality with a scanner: the code must read on the first try; otherwise goods may be rejected.

Is it difficult for staff to learn working with marking?

No, the system is fairly simple. Core actions for employees are:

- receive and verify DataMatrix codes in the “ASL Belgisi” system;
- connect the printer and print codes on labels;
- scan to ensure codes are readable;
- transmit product movement data into the system (often done automatically via integrated software).

With brief hands-on training, staff adopt these steps quickly — it’s enough to demonstrate how the printer and software work.

Bottom line: no special skills are required. With short training, any employee can print and verify codes without errors.

Can we print marking directly on packaging instead of on a label?

Yes, but there are important nuances:

Packaging material. Marking must be clear and durable. On cardboard or plastic, specialized printing is possible, but on thin or glossy surfaces the code may be poorly readable.

Printer type. Direct-to-packaging printing typically uses industrial equipment — thermal transfer or laser marking. Regular desktop thermal printers are not suitable.

Flexibility. It’s easier to change information on a label (e.g., shelf life, price, lot). With direct printing on packaging, updates are more difficult.

Bottom line: direct-to-pack printing is possible, but labels are more practical, safer and more reliable for most products.

How can we (Jicode) help businesses with marking?

Jicode helps implement marking quickly and without errors:

- Selecting equipment and supplies. We recommend the printer, ribbon and label that match your product.
- Configuring software. We connect the printer to the “ASL Belgisi” system and install the required printing software.
- Staff training. We run short practical sessions so your team can print and verify DataMatrix codes independently.
- Technical support. If issues arise with the printer or printing, we resolve them promptly on site or remotely.
- Legal and regulatory guidance. We explain the rules, monitor updates and help you stay compliant.

Bottom line: Jicode is not just a supplier — we are a partner who supports your business at every stage of the marking process to keep it simple and safe.